

AI POWERED LEGAL DOCUMENT GENERATOR

Automating the drafting of legal documents using Artificial Intelligence

Domain

Legal Technology + Artificial Intelligence

Approach

LLM + clause retrieval (RAG) + validation

Output

Review-ready DOCX and PDF

Problem Statement

Why ordinary users struggle with legal documents

Routine documents can become expensive when professional drafting is required.

Simple agreements may take days to prepare.

Users may sign terms they do not fully understand.

Missing clauses, incorrect names or dates can reduce document quality.

- Goal: provide a simple, affordable and reliable drafting workflow.

High Cost

Time-Consuming

Complex Language

Human Error

- Core gap: templates are rigid, while general AI tools may lack legal validation.

2/10

Objectives & Scope

What the project is designed to achieve

- Generate legal documents from simple user inputs
 - Draft clear, structured clauses with AI
 - Validate mandatory clauses
 - Reduce drafting time and routine cost
 - Export DOCX and PDF
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- Rental / lease agreement
 - NDA and affidavit
 - Offer letter and service agreement
 - Guided form, validation and summary
 - English language in the first version
 - Human review remains important

Objectives

Scope

3/10

Proposed System

A guided AI drafting platform	1. User Input
User selects a document type and answers simple guided questions.	2. AI Drafting
LLM creates clauses using user inputs and retrieved verified clauses.	3. Validation
Mandatory clauses, names, dates and conflicts are checked.	4. Review
User reviews and edits the generated draft.	5. Export
Final document is generated as DOCX or PDF.	4/10

System Architecture

Layered design for maintainability and scalability	Presentation Layer
Web forms • preview • document download	Application Layer
Backend API • authentication • workflow control	AI Layer
Template engine • LLM with retrieval • validation	Data Layer
Template library • legal knowledge base • user records	5/10

Technology Stack

Main technologies specified in the project report

React.js + Tailwind CSS

Python + FastAPI

LLM API + FAISS / ChromaDB

Jinja2 + python-docx + ReportLab

PostgreSQL

Frontend

Backend

AI / Retrieval

Documents

Database

Security / Deploy

Key Modules & Workflow

Six modules connect the user's input to the final document

Collect validated structured data

Choose document type and jurisdiction rules

Generate clauses from inputs

Fetch verified clauses and context

Flag missing or conflicting information

1. User Input

2. Template Selection

3. AI Drafting

4. Clause Retrieval

5. Validation & Review

6. Export & Storage

Supported Documents

Common documents targeted by the first version

Rent, deposit, duration

Confidential information, term

Role, salary, joining date

Name, address, statement

Principal, agent, powers

Rental / Lease Agreement

Non-Disclosure Agreement

Employment Offer Letter

Affidavit

Power of Attorney

Sale / Service Agreement

Security, Testing & Evaluation

Reliability and responsible use are built into the design

- HTTPS / TLS
- Encrypted storage
- JWT and role-based access
- Minimal data collection
- Audit activity logs
- Human review and disclaimer

- Valid rental inputs
- Missing-field validation
- Mandatory clause checks
- DOCX/PDF export
- Unauthorised access testing

- Clause completeness: 90%+
- Draft generation: under 30 seconds
- Readability for non-lawyers
- Positive user feedback

Security & Privacy

Testing

Evaluation Targets

9/10

Benefits, Future Enhancements & Conclusion

Benefits

- Faster routine drafting
- Lower routine cost
- Simple guided process
- Consistent structure
- Available online
- Easy to extend

Future Enhancements

- Tamil / Hindi support
- Voice input
- E-signature and e-stamping
- Contract risk analysis
- Lawyer review marketplace
- Mobile apps

Conclusion

AI can reduce repetitive drafting work by combining an LLM with verified clauses and automatic validation. It is a drafting aid—not a replacement for legal expertise. Important documents should be reviewed by a qualified professional.

10/10